Question 1 - Parts Order Form

a. Assumptions

I am treating each customer number as the identifier for a customer, so it determines the customer's name and type. I am also treating each part number as the identifier for a part, so it determines the part name, type, cage code, and unit price. The cage code identifies the shelf or cage where inventory is stored; more than one part can be stored in the same cage.

Since the form does not include an order number, and the assignment says not to add fields, I am using **customerNo**, **date**, and **time** together to identify an order. This assumes that the same customer does not place two separate orders at the exact same date and time. Each order is handled by one employee, and employees can help any customer. I also assume a part appears only once per order, with the total quantity listed on that order line.

b. Normalization Process

Original Table (UNF)

The original form mixes order and customer information with a repeated list of parts. I use **customerNo**, **date**, and **time** to identify the order, but the part fields still contain several values in one row.

customerNo	**date**	**time**	customerName	customerType	employee	partNo	partName	type	cageCode	quantityOrdered	unitPrice
HG54587	7/1/2024	10:30 AM	Jeff Peterson	Consumer	D Harrison	10654; 10456; 10776; 10657	Float Control; Modulator; Hose Assembly; Float Assembly	Plumbing; Electrical; Plumbing; Plumbing	G413; H433; G413; G413	4; 3; 7; 5	12; 7; 9; 10

The part values line up by position across the part columns. This table is not in 1NF yet because those columns are repeating groups rather than single values.

First Normal Form (1NF)

For 1NF, I put each ordered part on its own row. At that point, the primary key is **customerNo** + **date** + **time** + **partNo**.

customerNo	**date**	**time**	**partNo**	customerName	customerType	employee	partName	type	cageCode	quantityOrdered	unitPrice
HG54587	7/1/2024	10:30 AM	10654	Jeff Peterson	Consumer	D Harrison	Float Control	Plumbing	G413	4	12
HG54587	7/1/2024	10:30 AM	10456	Jeff Peterson	Consumer	D Harrison	Modulator	Electrical	H433	3	7
HG54587	7/1/2024	10:30 AM	10776	Jeff Peterson	Consumer	D Harrison	Hose Assembly	Plumbing	G413	7	9
HG54587	7/1/2024	10:30 AM	10657	Jeff Peterson	Consumer	D Harrison	Float Assembly	Plumbing	G413	5	10

Now each cell has one value, so the table is in 1NF. There are still partial dependencies, though. customerName and customerType depend only on customerNo. partName, type, cageCode, and unitPrice depend only on partNo. employee depends on the order information: customerNo + date + time. quantityOrdered is the only non-key field here that needs the full key, since it describes a specific part within a specific order.

Second Normal Form (2NF)

For 2NF, I separated the fields that depended on only part of the key. customer data is stored once per customer, part data once per part, order-level data once per order, and quantity data once per ordered part.

customer

customerNo	customerName	customerType
HG54587	Jeff Peterson	Consumer

part

partNo	partName	type	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order

customerNo	**date**	**time**	employee
HG54587	7/1/2024	10:30 AM	D Harrison

order_part

customerNo	**date**	**time**	**partNo**	quantityOrdered
HG54587	7/1/2024	10:30 AM	10654	4
HG54587	7/1/2024	10:30 AM	10456	3
HG54587	7/1/2024	10:30 AM	10776	7
HG54587	7/1/2024	10:30 AM	10657	5

With those splits, the partial dependencies are removed, so these relations are in 2NF.

Third Normal Form (3NF)

For this question, the 2NF relations are already in 3NF. No non-key attribute determines another non-key attribute, so there is no transitive dependency left to split out. The final relations are below; primary keys are bold and underlined, and foreign keys are italicized.

customer

customerNo	customerName	customerType
HG54587	Jeff Peterson	Consumer

part

partNo	partName	type	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12

10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order

customerNo	**date**	**time**	employee
HG54587	7/1/2024	10:30 AM	D Harrison

order_part

customerNo	***date***	***time***	***partNo***	quantityOrdered
HG54587	7/1/2024	10:30 AM	10654	4
HG54587	7/1/2024	10:30 AM	10456	3
HG54587	7/1/2024	10:30 AM	10776	7
HG54587	7/1/2024	10:30 AM	10657	5

c. Primary and Foreign Keys

The original form is identified by **customerNo** + **date** + **time**. Once the parts are expanded into separate rows in 1NF, the key becomes **customerNo** + **date** + **time** + **partNo**. The keys for the final relations are listed below.

Final Relation	Primary Key	Foreign Keys
customer	**customerNo**	None
part	**partNo**	None
order	***customerNo*** + **date** + **time**	*customerNo* references customer(customerNo)
order_part	***customerNo*** + ***date*** + ***time*** + ***partNo***	*(customerNo, date, time)* references order(customerNo, date, time); *partNo* references part(partNo)

In order_part, customerNo, date, and time work together as the composite foreign key back to order. partNo is a separate foreign key back to part.

d. Relation Names

The final 3NF relation names are customer, part, order, and order_part.

Question 2 - Therapist Appointment Data

a. Assumptions

I am treating staffNo as the identifier for a therapist, so it determines therapistName. I am treating patNo the same way for a patient, so it determines patName. For appointments, I am using **patNo**, **date**, and **time** as the key. That allows a patient to have more than one appointment on the same day, as long as the times are different.

Each appointment is with one therapist. A therapist can work at different branches on different days, but only at one branch on a given day. Because of that, **staffNo** and **date** together determine branchNo.

b. Normalization Process

Original Table (1NF)

The original table already has single values in its cells, so it is in 1NF. The primary key is **patNo** + **date** + **time**.

patNo	**date**	**time**	patName	staffNo	therapistName	branchNo
P100	9/12/2022	10:00	Lily White	S1011	Fred Smith	M15
P105	9/12/2022	12:00	Jill Baker	S1011	Fred Smith	M15
P108	9/12/2022	10:00	Andy McKee	S1024	Heidi Pierce	Q10
P108	9/14/2022	14:00	Andy McKee	S1024	Heidi Pierce	Q10
P105	9/14/2022	16:30	Jill Baker	S1032	Richard Levin	M15
P110	9/15/2022	18:00	Jimmy Winter	S1032	Richard Levin	B13

Even though the table is in 1NF, it still repeats data. patName depends only on patNo, and therapistName depends only on staffNo. branchNo depends on staffNo and date, since that combination tells us where the therapist is working that day.

Second Normal Form (2NF)

For 2NF, I first separate the patient information. patient data is stored once per patient, while the appointment-level data stays in a separate table.

patient

patNo	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

appointment_detail

patNo	**date**	**time**	staffNo	therapistName	branchNo
P100	9/12/2022	10:00	S1011	Fred Smith	M15
P105	9/12/2022	12:00	S1011	Fred Smith	M15
P108	9/12/2022	10:00	S1024	Heidi Pierce	Q10
P108	9/14/2022	14:00	S1024	Heidi Pierce	Q10
P105	9/14/2022	16:30	S1032	Richard Levin	M15
P110	9/15/2022	18:00	S1032	Richard Levin	B13

This removes the partial dependency patNo -> patName. There are still dependencies between non-key fields, though: staffNo determines therapistName, and staffNo + date determines branchNo. Those need to be split out for 3NF.

Third Normal Form (3NF)

For 3NF, I separate therapist information and the therapist's daily branch schedule from the appointment itself. The final relations are below; primary keys are bold and underlined, and foreign keys are italicized.

patient

patNo	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

therapist

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

therapist_schedule

staffNo	**date**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

appointment

patNo	***date***	**time**	*staffNo*
P100	9/12/2022	10:00	S1011
P105	9/12/2022	12:00	S1011
P108	9/12/2022	10:00	S1024
P108	9/14/2022	14:00	S1024
P105	9/14/2022	16:30	S1032

P110	9/15/2022	18:00	S1032

c. Primary and Foreign Keys

The original 1NF table is identified by **patNo** + **date** + **time**. The final 3NF keys are listed below.

Final Relation	Primary Key	Foreign Keys
patient	**patNo**	None
therapist	**staffNo**	None
therapist_schedule	***staffNo*** + **date**	*staffNo* references therapist(staffNo)
appointment	***patNo*** + ***date*** + **time**	*patNo* references patient(patNo); *(staffNo, date)* references therapist_schedule(staffNo, date)

In appointment, staffNo and date together point back to the therapist's branch assignment for that date.

d. Relation Names

The final 3NF relation names are patient, therapist, therapist_schedule, and appointment.

Question 3 - Employee Contract Hours

a. Assumptions

I am treating eNo as the identifier for an employee, so it determines eName. contractNo identifies a contract. Each contract applies to one event, while one event can have multiple contracts for different service needs. eventNo determines the event location.

I am treating hours as the number of hours one employee worked on one contract. That means the original table is identified by **eNo** + **contractNo**.

b. Normalization Process

Original Table (1NF)

The original table already has single values in each cell, so it is in 1NF. The primary key is **eNo** + **contractNo**.

eNo	**contractNo**	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

The table is in 1NF, but it still has partial dependencies. eName depends only on eNo. eventNo depends only on contractNo, since each contract is tied to one event. eventLoc depends on eventNo, so the event location is also being repeated through the contract data.

Second Normal Form (2NF)

For 2NF, I separate the fields that depend on only part of the composite key. employee information is stored once per employee, contract information once per contract, and hours stays with the employee-contract combination.

employee

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

contract

contractNo	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

work_assignment

eNo	**contractNo**	hours
1135	C1024	16
1057	C1024	24

1068	C1025	28
1135	C1025	15
1135	C1026	10

This removes the partial dependencies. contract still has a transitive dependency, though: contractNo determines eventNo, and eventNo determines eventLoc. That needs to be separated for 3NF.

Third Normal Form (3NF)

For 3NF, I separate the event location into its own relation. The final relations are below; primary keys are bold and underlined, and foreign keys are italicized.

employee

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

event

eventNo	eventLoc
H25	Queens

H4	Yonkers

contract

<u>contractNo</u>	*eventNo*
C1024	H25
C1025	H4
C1026	H25

work_assignment

<u>eNo</u>	***<u>contractNo</u>***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

c. Primary and Foreign Keys

The original 1NF table is identified by **eNo** + **contractNo**. The final 3NF keys are listed below.

Final Relation	Primary Key	Foreign Keys
employee	**eNo**	None
event	**eventNo**	None
contract	**contractNo**	*eventNo* references event(eventNo)
work_assignment	***eNo*** + ***contractNo***	*eNo* references employee(eNo); *contractNo* references contract(contractNo)

d. Relation Names

The final 3NF relation names are employee, event, contract, and work_assignment.